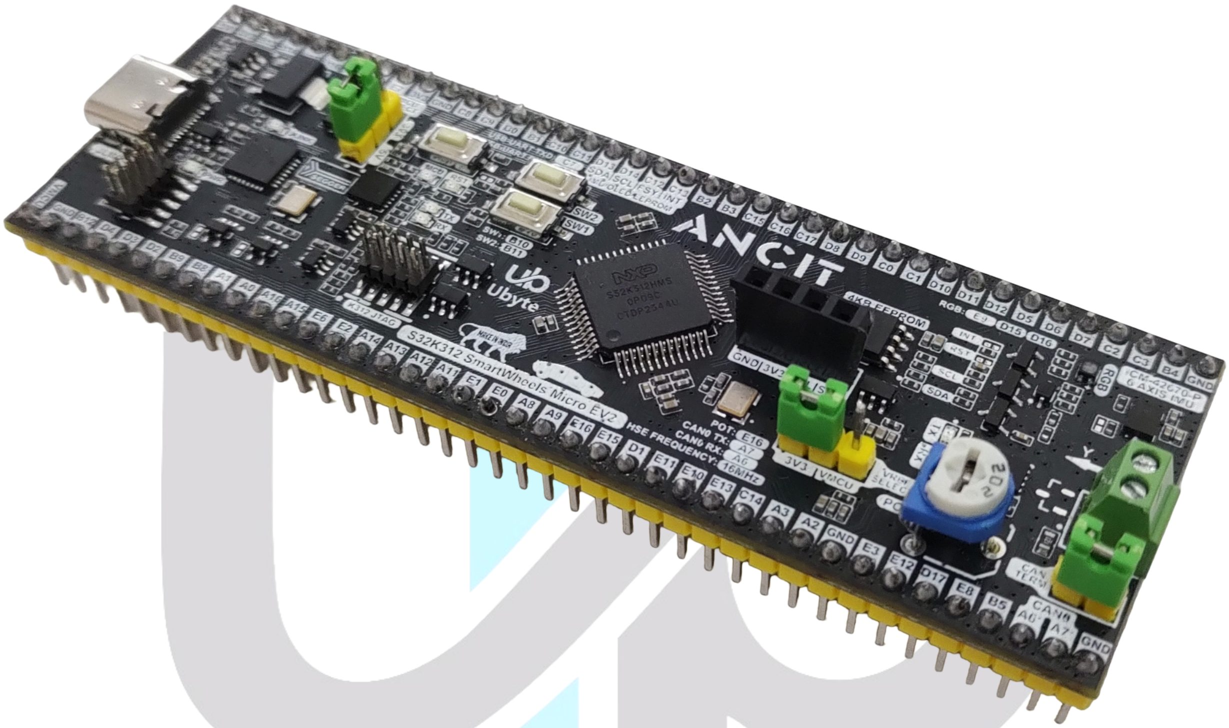
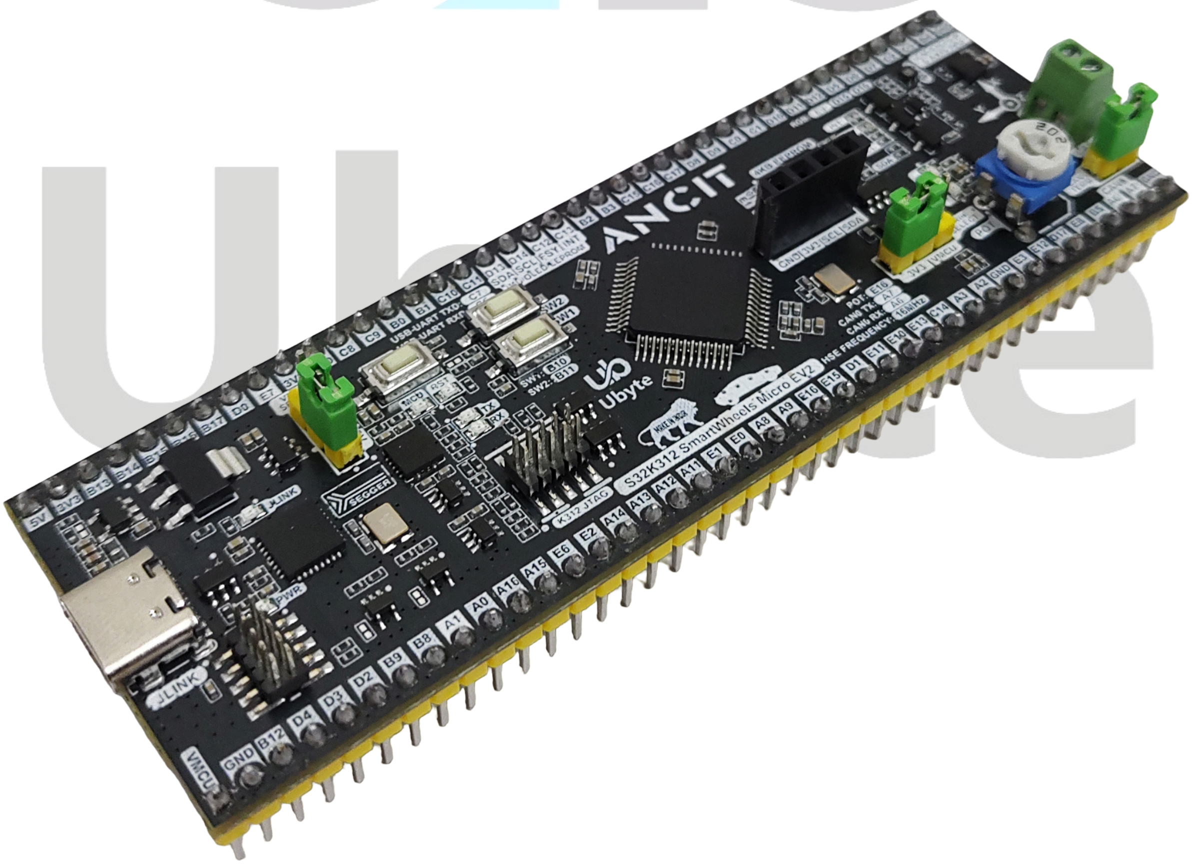
S32K312 SmartWheels Micro EV2
MATLAB MBDT User Manuel





NXP Model-Based Design Toolbox for S32K3: Installation and Configuration Guide

Contents

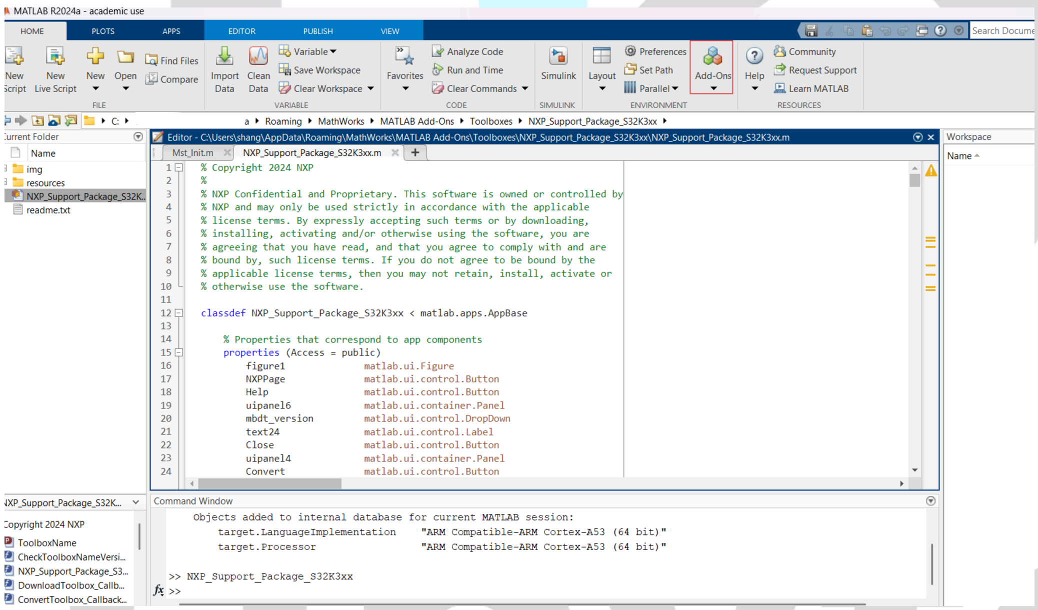
1. Introduction

This document outlines the process for installing the Model-Based Design Toolbox (MBDT), which is the first step in setting up and running automatic C code generation from MATLAB/Simulink for NXP's embedded target processors and development boards. The NXP Support Package S32K3 is a graphical user interface guide designed to assist with the download, installation, and activation of the MBDT for S32K3.

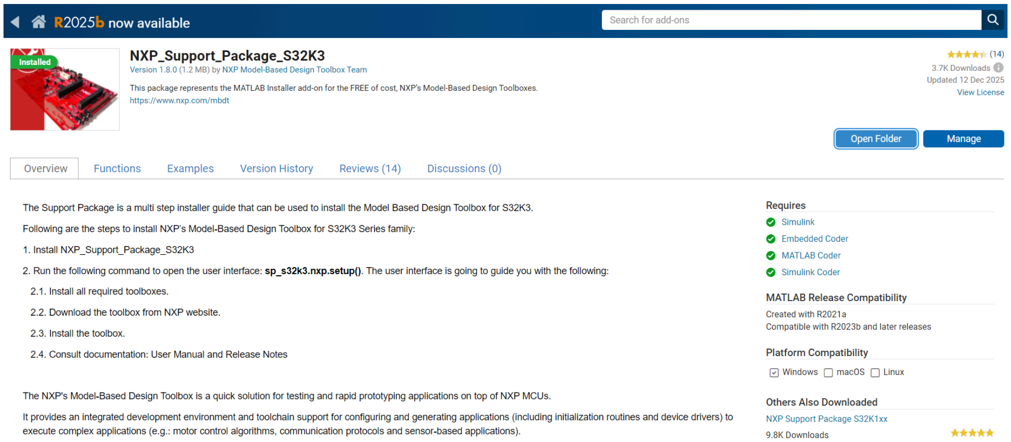
2. Installation Procedure

2.1 Installing the Support Package

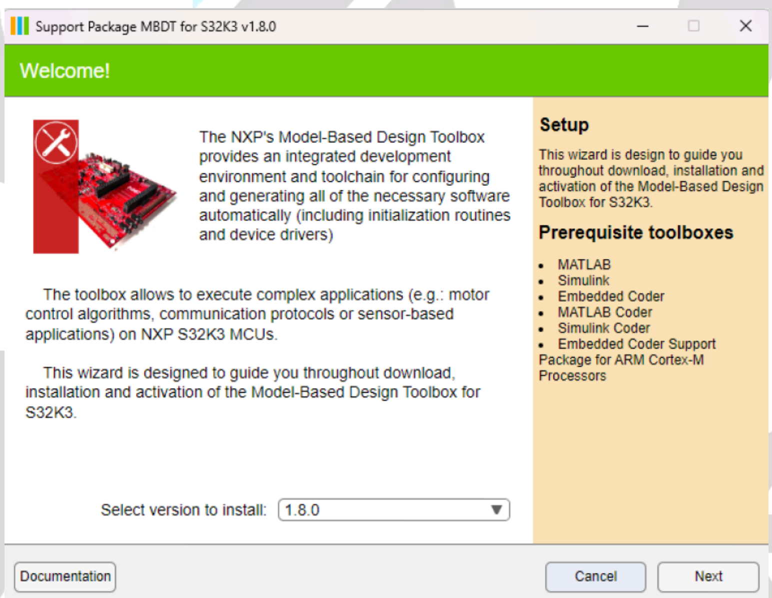
1. **Navigate to Add-Ons:** Open MATLAB and go to **MATLAB Add-Ons**



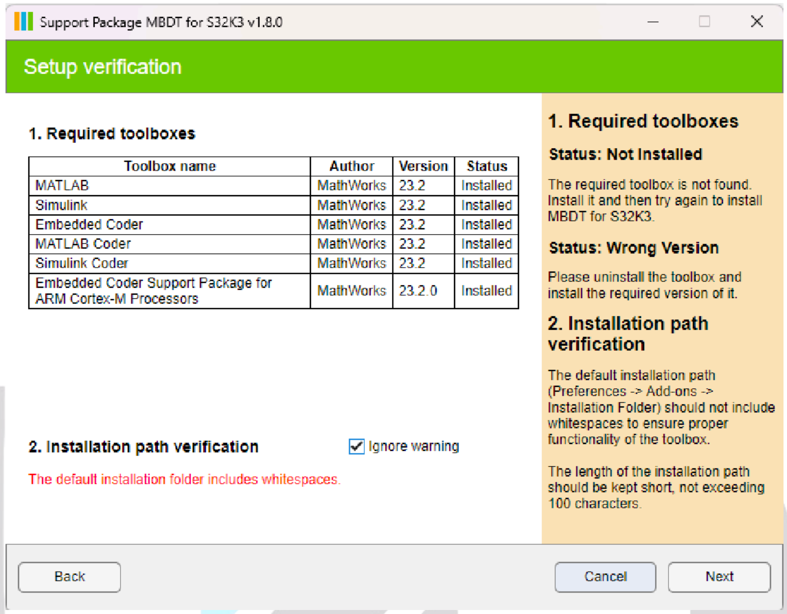
2. **Search and Install:** Go to the Add-On Explorer, search for "NXP Support Package S32K3," and install it.
S32K3"https://nl.mathworks.com/matlabcentral/fileexchange/101749-nxp_support_package_s32k3



3. **Launch Installer:** Run the command sp_s32k3.nxp.setup() to open the user interface if it does not open automatically.



4. **Version Selection:** The package acts as a GUI guide; select the toolbox version to install (e.g., 1.8.0) and press **Next**.
5. **Software Verification:** Check that all required software (Simulink, Embedded Coder, etc.) is installed and press **Next**.

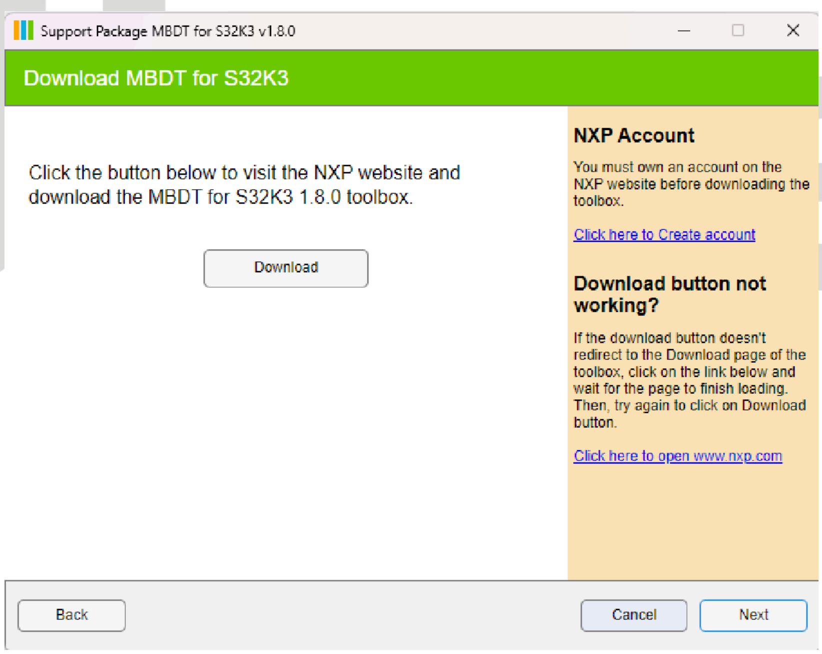


6. Download the SW32_MBDT_S32K3_x.x.x_DYYMM.mltbx file.

   Note: If the downloaded file has a .zip extension instead of .mltbx, the next step will help convert it to the correct format

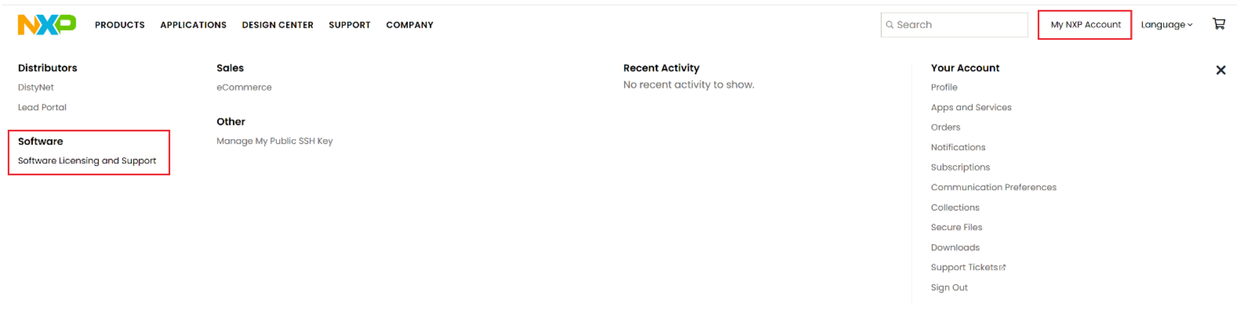
2.2 Finalizing Installation

1. Return to the **NXP Support Package for S32K3** installer and press the **Install** button.

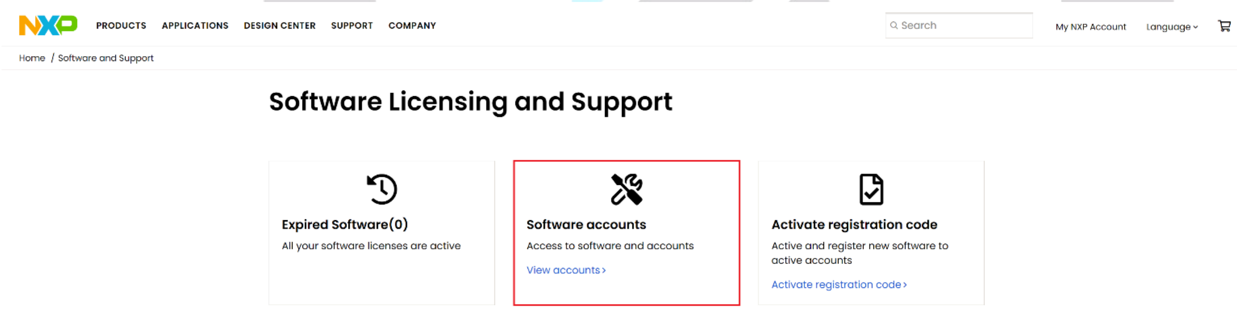


2. Review and accept the Terms and Conditions by pressing **I Accept**

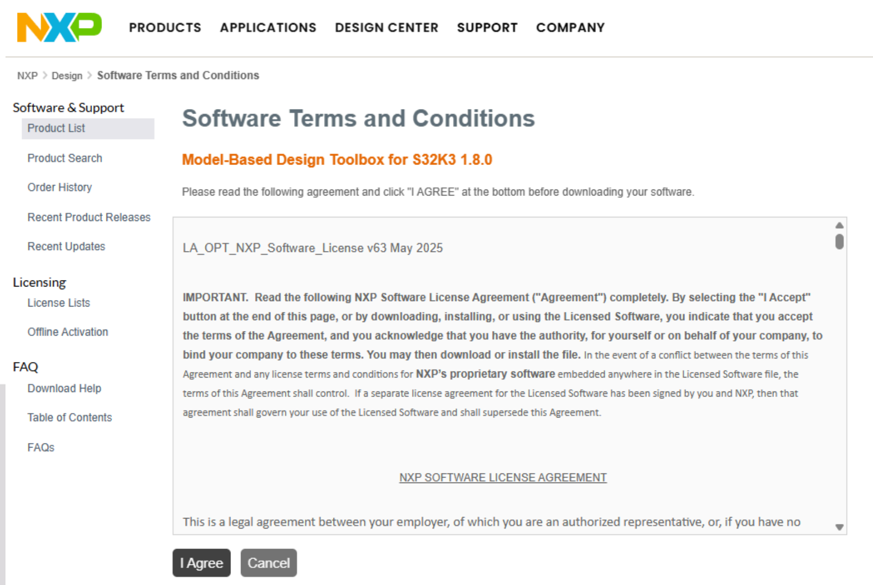
Note: If the page is not displayed as below, please go to the nxp.com website, log into your account (or make one for free). Acces **My NXP Account** -> **Software Licensing and Support**:



3. For the next step go to **Software accounts** -> **View accounts**

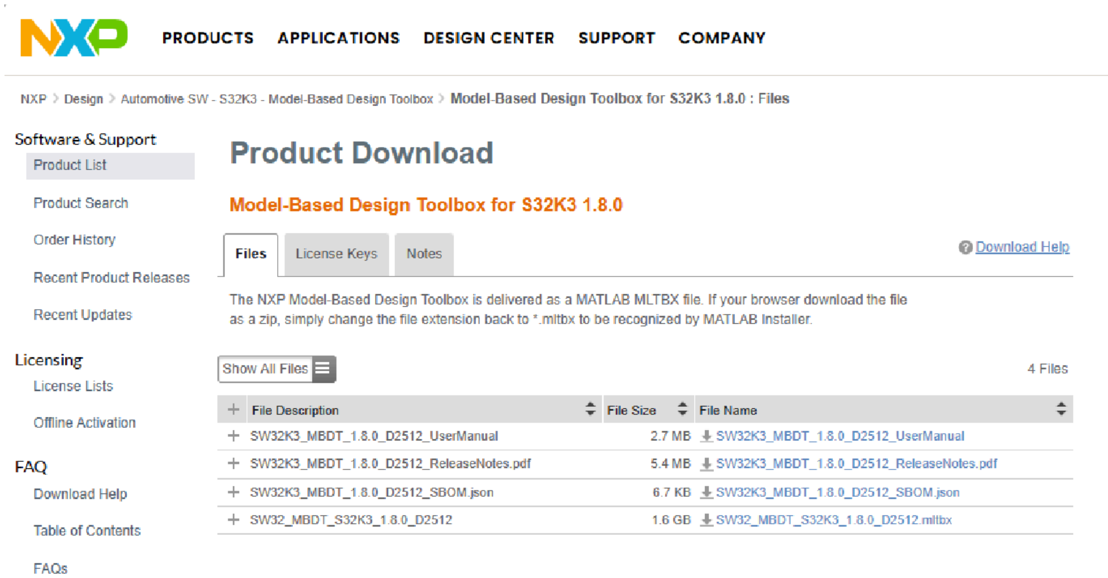


4. You will find a Product list on the next page from which you should select **Automotive SW – S32K3 Standard Software.** Go to **Automotive SW - S32K3 - Model-Based Design Toolbox** and select the latest release available.

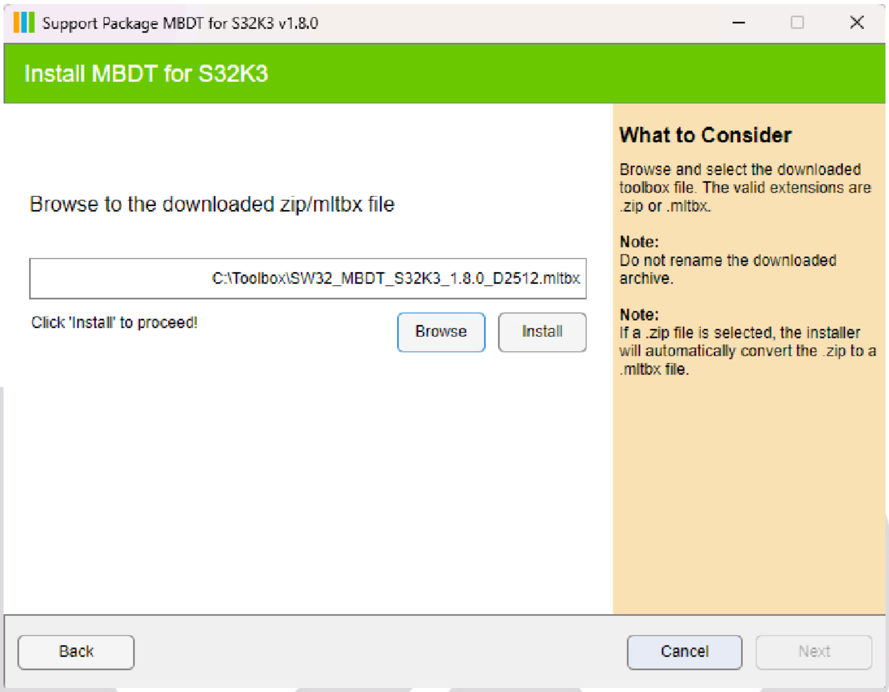


5. Download the **SW32_MBDT_S32K3_x.x.x_DYYMM.mltbx** file.

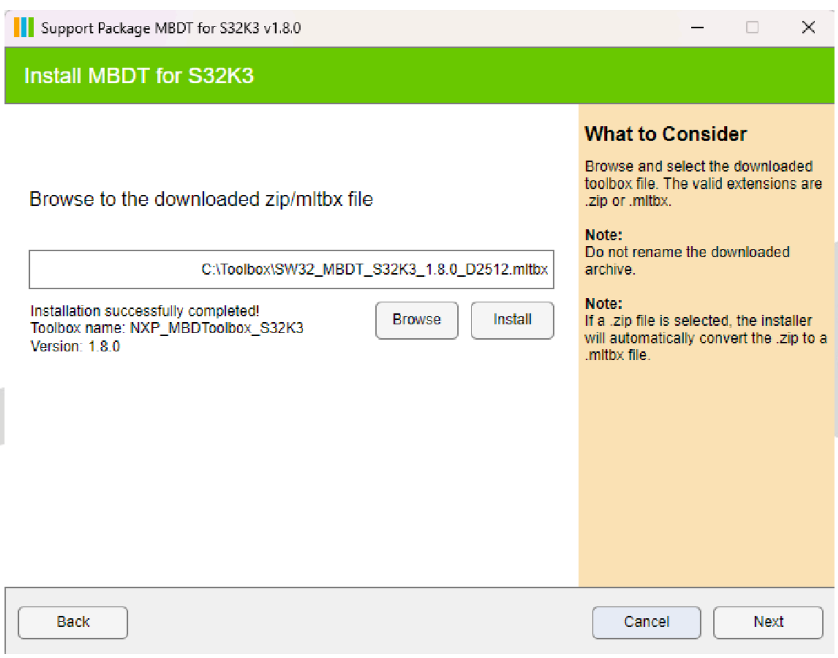
Note: The downloaded file has the **.zip** extension instead of **.mltbx**. The next step helps to convert to the right format. Please note that in the screenshot below the older naming convention for the toolbox is used. Please download the **SW32_MBDT_S32K3_x.x.x_DYYMM.mltbx** file.



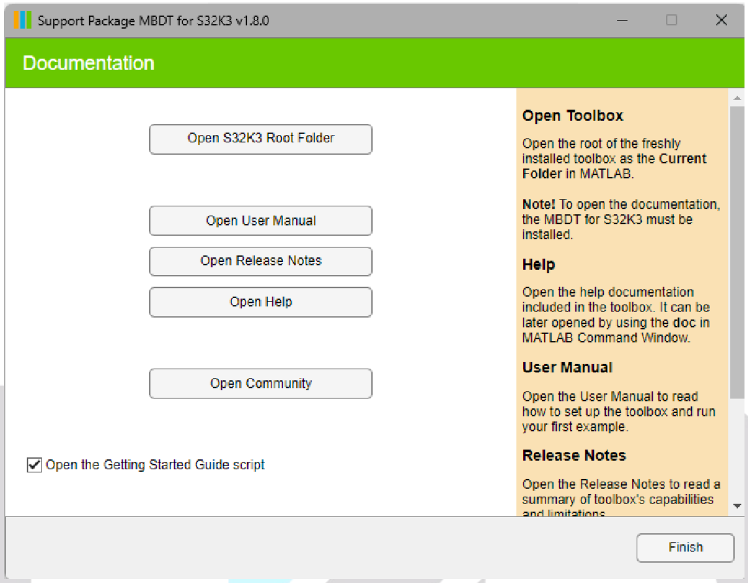
Go back to **NXP Support Package for S32K3** and press the **Install** button. Review the Terms and Conditions as you scroll down, and press the **I Accept** button. This action starts MBDT for S32K3 Toolbox installation process.



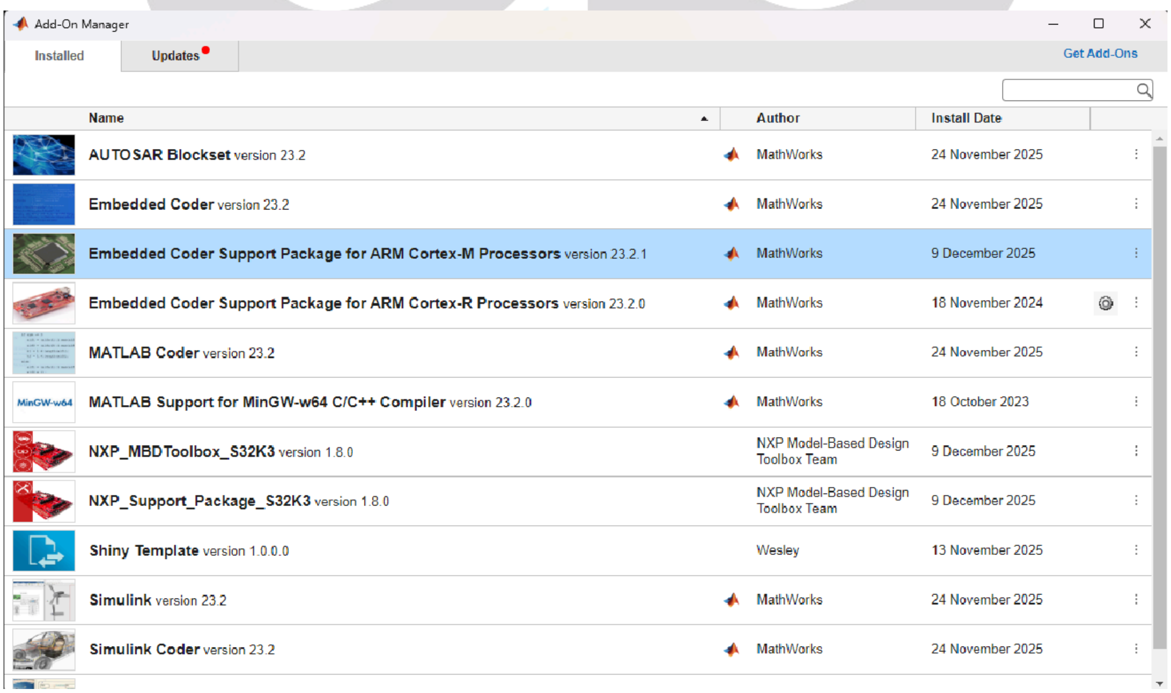
Installing the toolbox might take several minutes. Once the toolbox is successfully installed, press the **Next** button.



Open or enable the **Open Getting Started Guide** to complete the setup



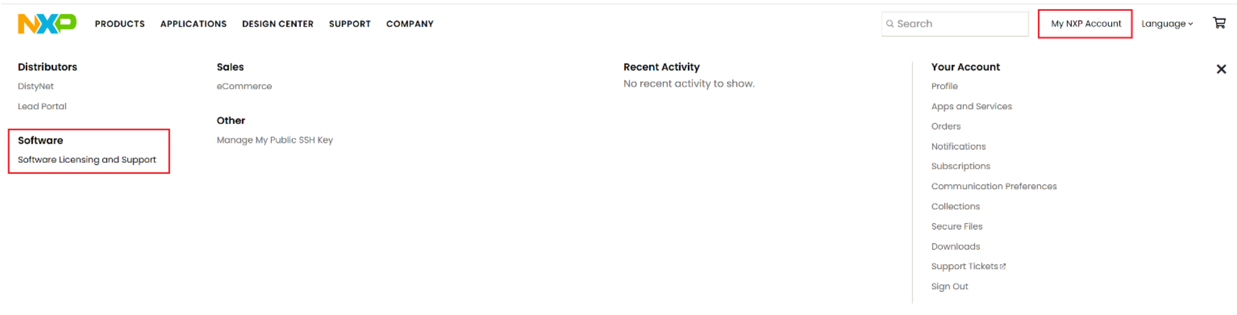
This mechanism requires users to install the Embedded Coder Support Package for ARM Cortex-M Processor as a prerequisite.



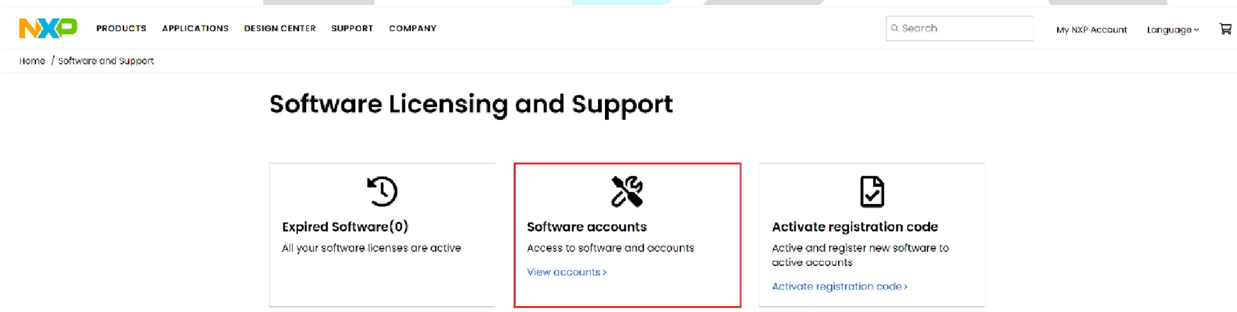
2.3 Installing EB Tresos

Model-Based Design Toolbox for S32K3 provides support for 2 external configuration tools – NXP's S32 Configuration Tools (which is made available directly through the mltbx installer and requires no additional installation) and EB tresos. To install this product, you will need to follow these instructions:

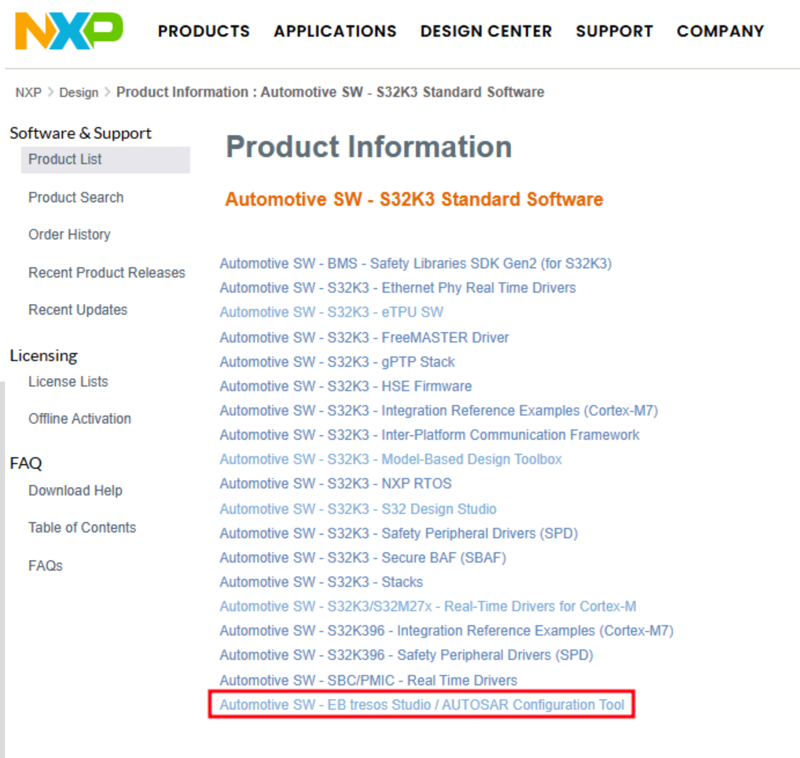
Go to the nxp.com website, log into your account (or make one for free). Then go to **My NXP Account** -> **Software Licensing and Support**:



For the next step go to **Software accounts** -> **View accounts:**

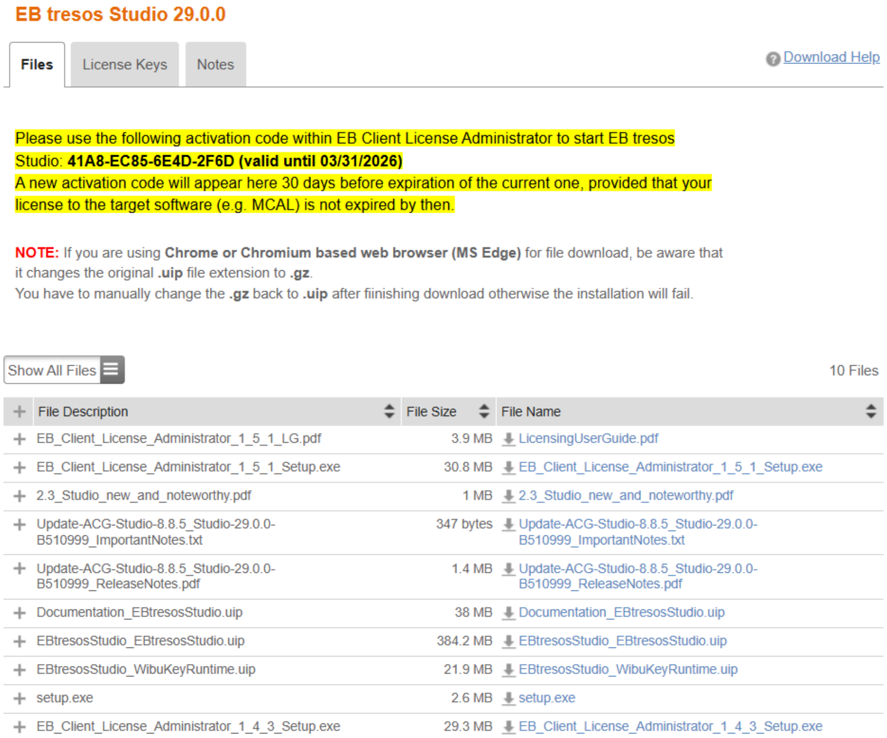


You will find a Product list on the next page from which you should select **Automotive SW – S32K3 Standard Software.** From the Product Information page, you should find the item names **Automotive SW – EB tresos Studio / AUTOSAR Configuration Tool**.



click here:https://nxp.flexnetoperations.com/control/frse/download?element=13469337

Download the files below



EB tresos Studio 29.0.0

Files | License Keys | Notes

Download Help

Please use the following activation code within EB Client License Administrator to start EB tresos Studio: **41A8-EC85-6E4D-2F6D (valid until 03/31/2026)**
A new activation code will appear here 30 days before expiration of the current one, provided that your license to the target software (e.g. MCAL) is not expired by then.

NOTE: If you are using **Chrome or Chromium based web browser (MS Edge)** for file download, be aware that it changes the original **.uip** file extension to **.gz**.
You have to manually change the **.gz** back to **.uip** after fiinishing download otherwise the installation will fail.

Show All Files

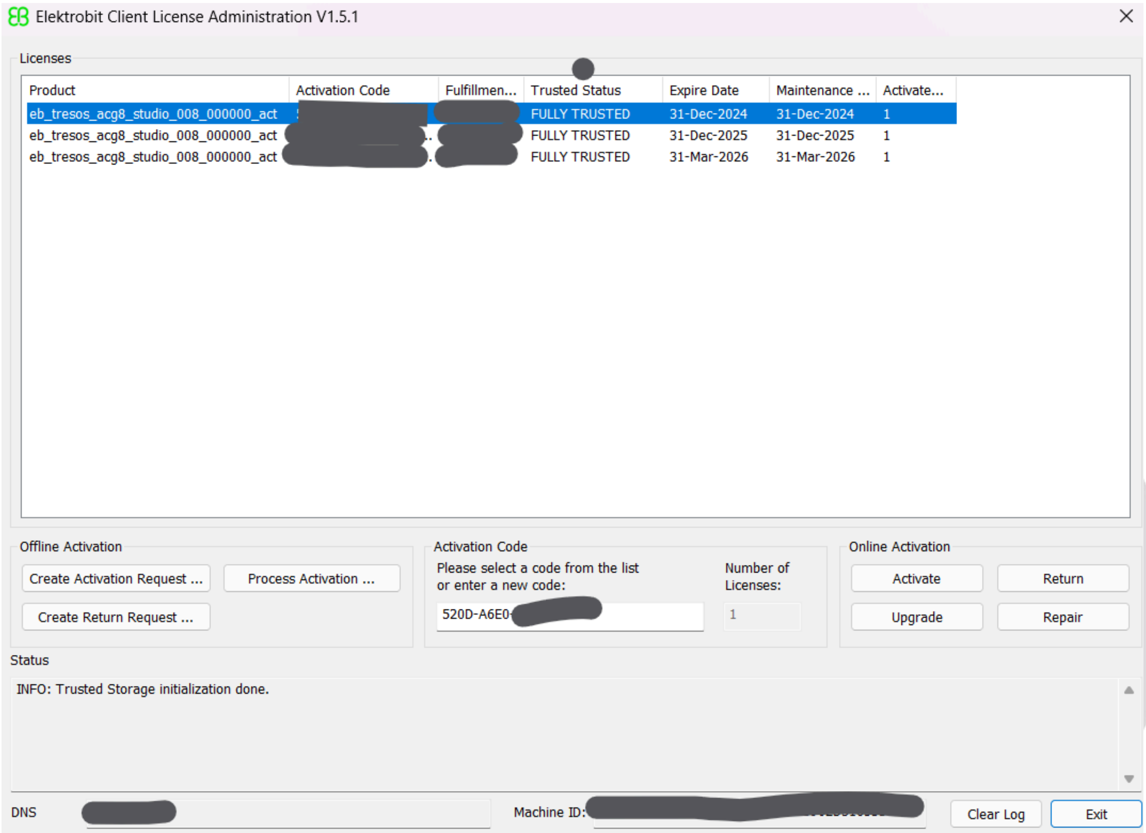
10 Files

File Description	File Size	File Name
EB_Client_License_Administrator_1_5_1_LG.pdf	3.9 MB	LicensingUserGuide.pdf
EB_Client_License_Administrator_1_5_1_Setup.exe	30.8 MB	EB_Client_License_Administrator_1_5_1_Setup.exe
2.3_Studio_new_and_noteworthy.pdf	1 MB	2.3_Studio_new_and_noteworthy.pdf
Update-ACG-Studio-8.8.5_Studio-29.0.0-B510999_ImportantNotes.txt	347 bytes	Update-ACG-Studio-8.8.5_Studio-29.0.0-B510999_ImportantNotes.txt
Update-ACG-Studio-8.8.5_Studio-29.0.0-B510999_ReleaseNotes.pdf	1.4 MB	Update-ACG-Studio-8.8.5_Studio-29.0.0-B510999_ReleaseNotes.pdf
Documentation_EBtresosStudio.uip	38 MB	Documentation_EBtresosStudio.uip
EBtresosStudio_EBtresosStudio.uip	384.2 MB	EBtresosStudio_EBtresosStudio.uip
EBtresosStudio_WibuKeyRuntime.uip	21.9 MB	EBtresosStudio_WibuKeyRuntime.uip
setup.exe	2.6 MB	setup.exe
EB_Client_License_Administrator_1_4_3_Setup.exe	29.3 MB	EB_Client_License_Administrator_1_4_3_Setup.exe

Download the setup.exe and License Administrator. And follow the Note mentioned in the image.

Install the setup by following the Note.

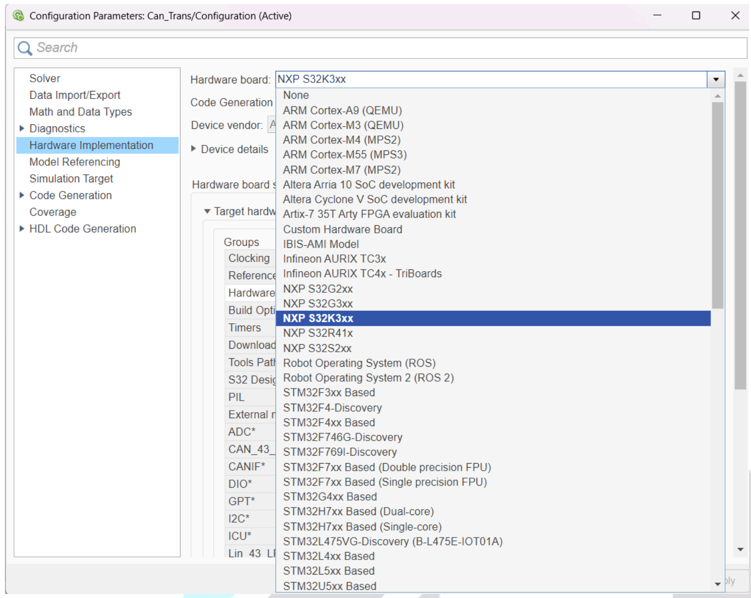
Install the License Administrator and enter the Activation key shown in the Image (Yellow Highlighter)



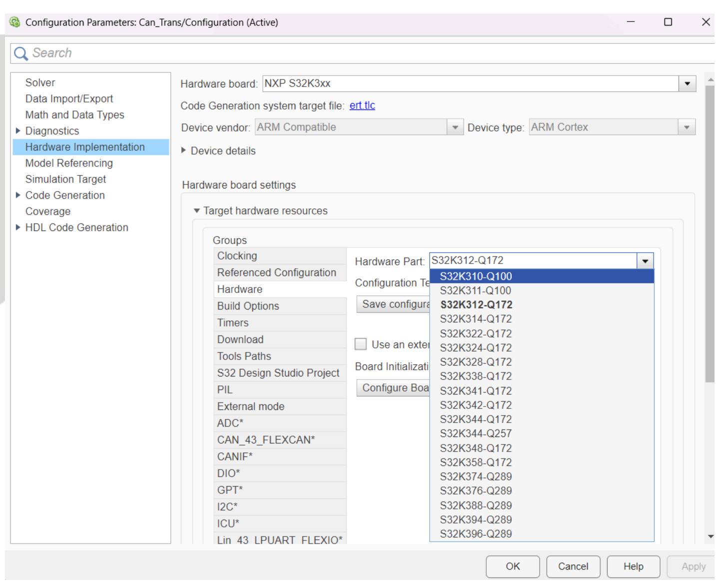
3. Simulink Model Configuration

3.1 Hardware Board Setup

- Target Selection: In Simulink, open the Configuration Parameters.



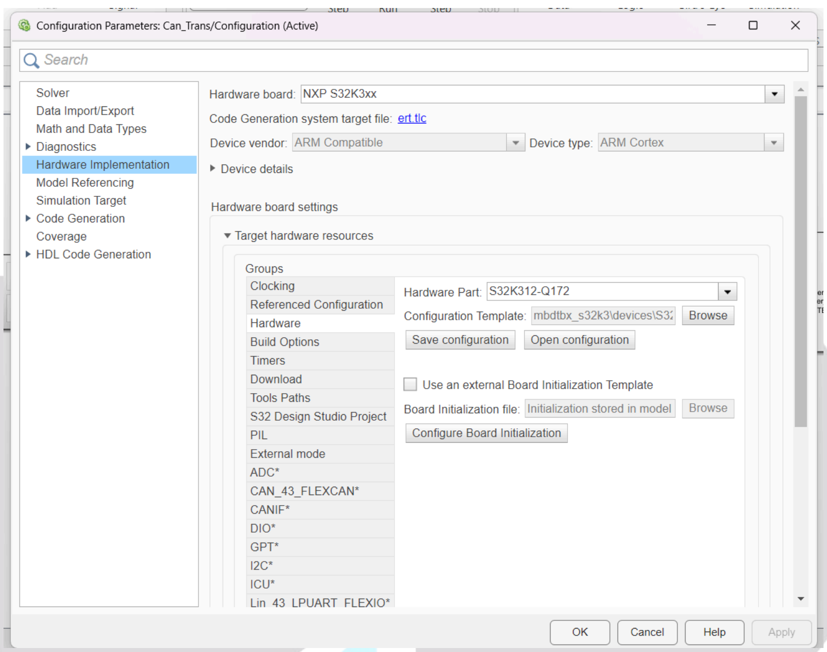
- Navigate to **Hardware Implementation**.
- In the **Hardware Board** dropdown, select **NXP S32K3xx**.
- Expand the **Target hardware resources** tab to access the **Hardware** group.
- Under **Hardware Part**, select the specific chip, such as **S32K312 – Q172**



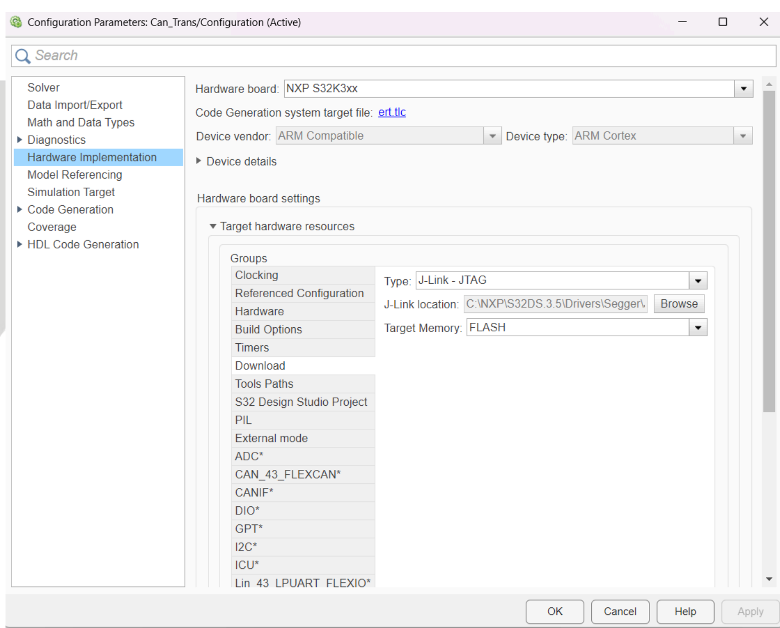
3.2 Flashing and Toolchain Configuration

1. **JTAG Setup:**
   1. Go to **Target hardware resources** > **Download**.

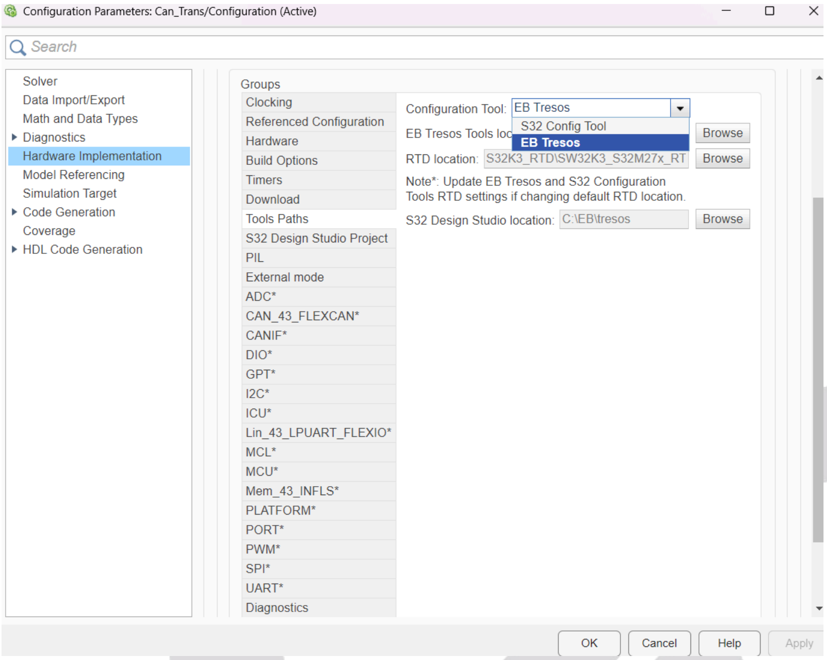
2. Select **JTAG** as the interface.
3. Browse and select the J-Link application path (e.g., C:\NXP\S32DS.3.5\Drivers\Segger\).



2. **Configuration Tool Path:**
   - Go to **Target hardware resources** > **Tools Paths**.

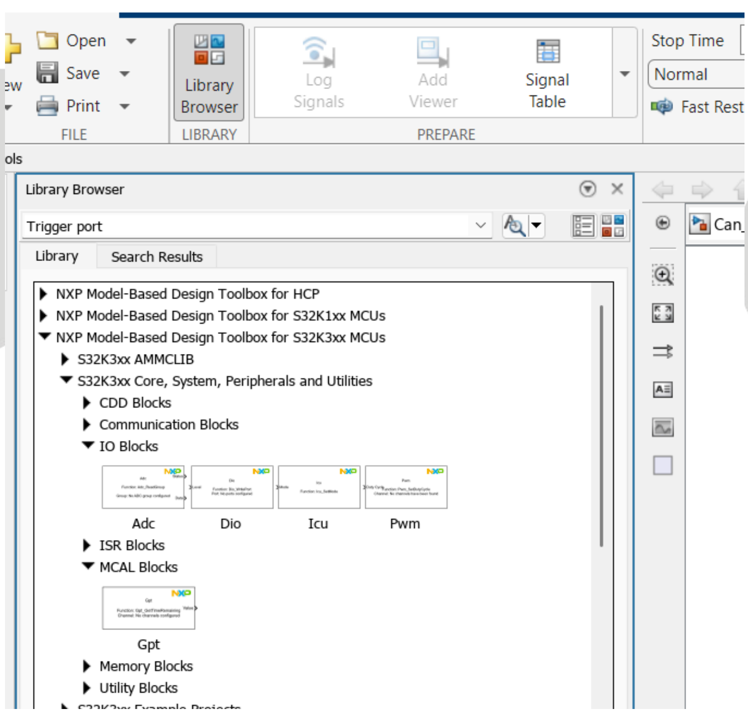


- Select **EB Tresos** or **S32 Config Tool** based on preference.



- Selecting the tool will generally automatically populate the paths

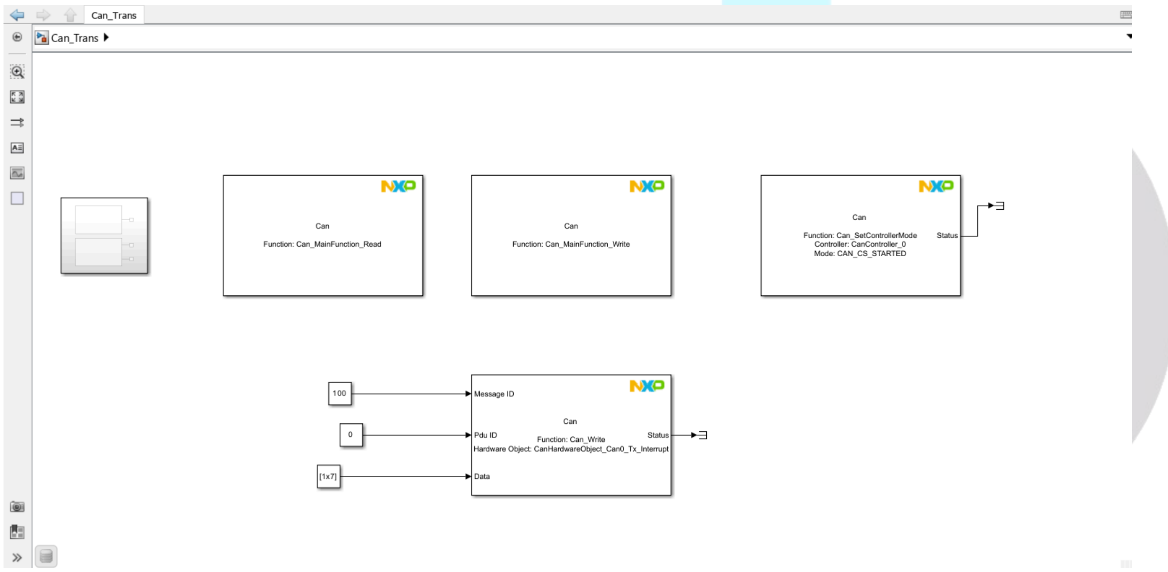
Library Access: Open the **Library Browser** and locate the **NXP Model-Based Design Toolbox for S32K3xx MCUs Library**.



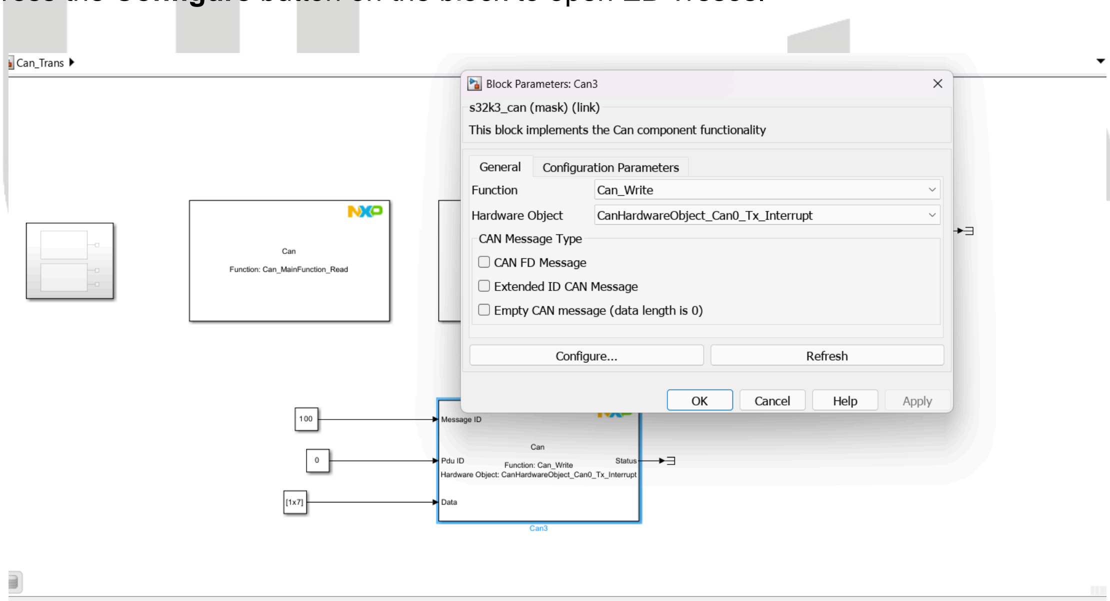
4. Implementation Examples

4.1 CAN TRANSMIT

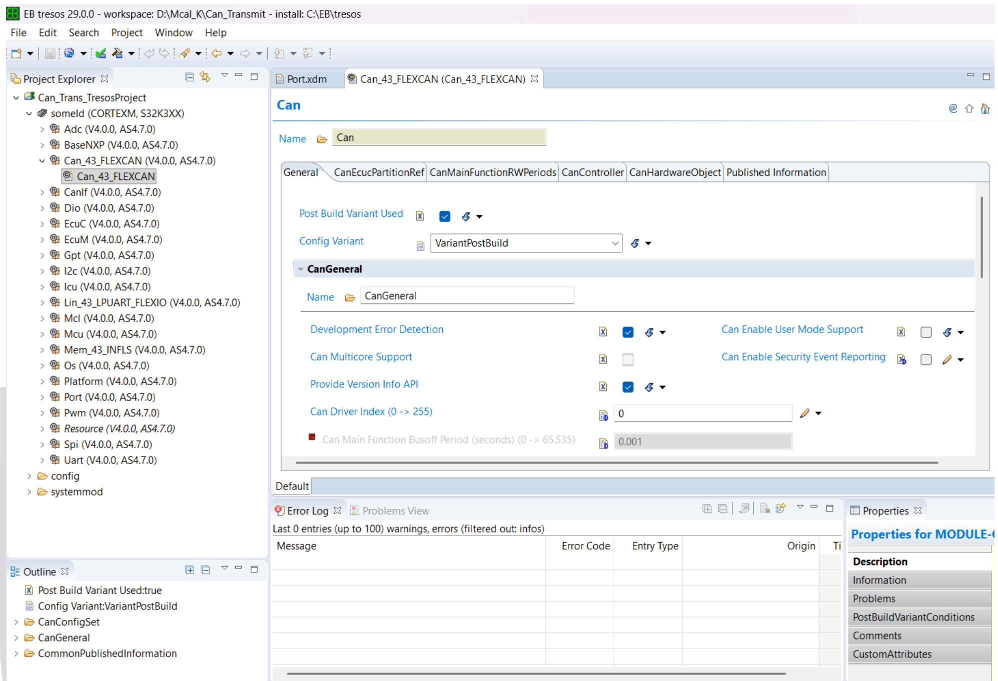
1. **Block Selection:** Drag the **CAN TRANSMIT** block into your model.
2. **Block Configuration:**
   - Set the **Can ID** (e.g., 100).
   - Verify data transmission using a CAN Analyzer Tool

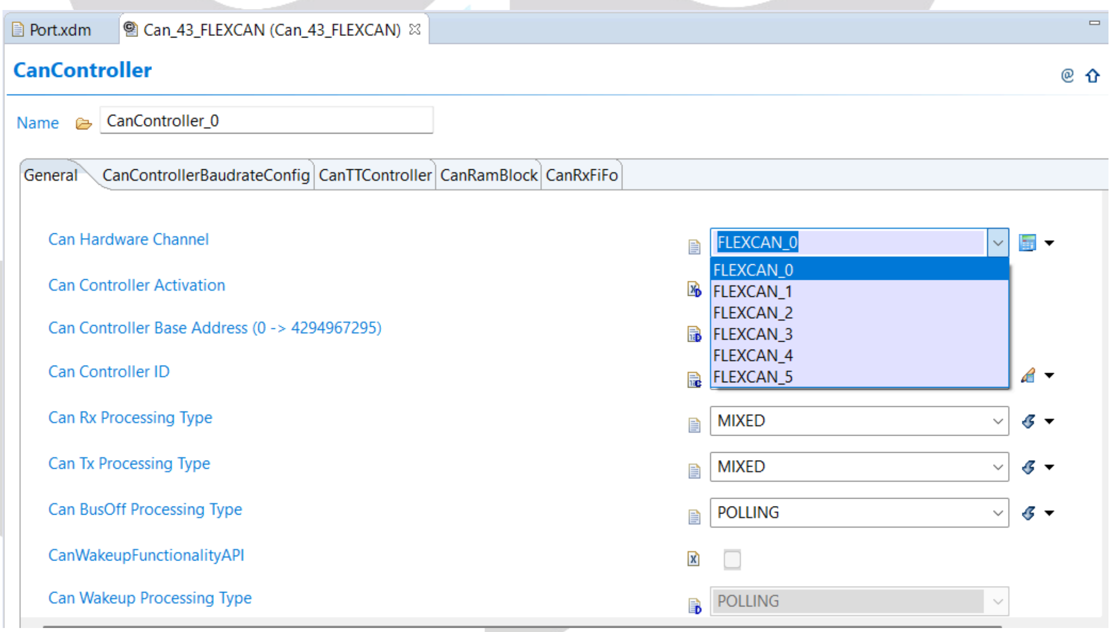


3. **EB Tresos Configuration:**
   - Press the **Configure** button on the block to open EB Tresos.

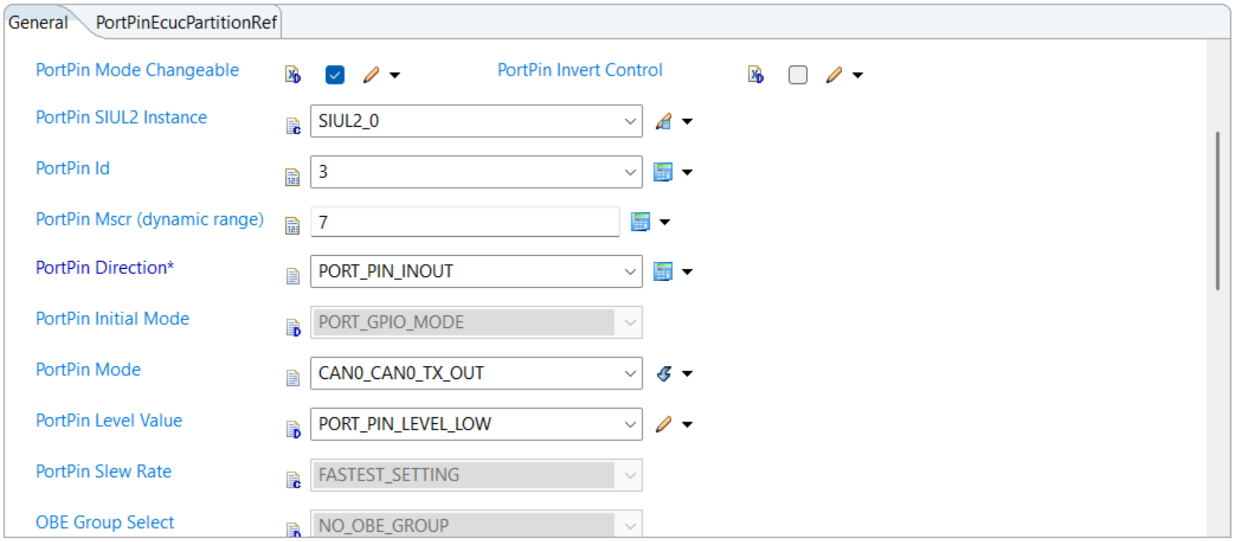


   - Navigate to **CanController** and select the PIN according to the Board Datasheet.

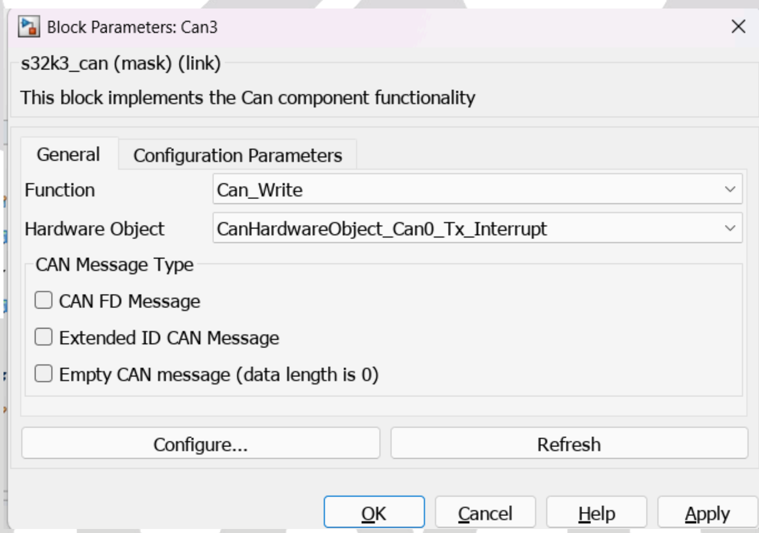




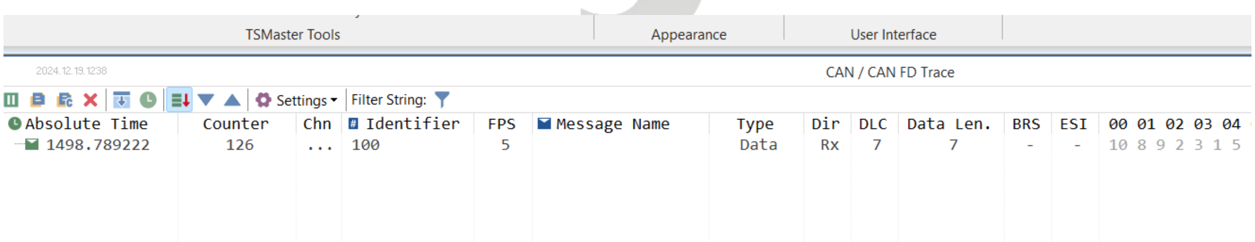
- Go to **Port container** > **CAN** > **Port pins** to configure the pins as per the datasheet.



4. **Build:** Update the code and build the model

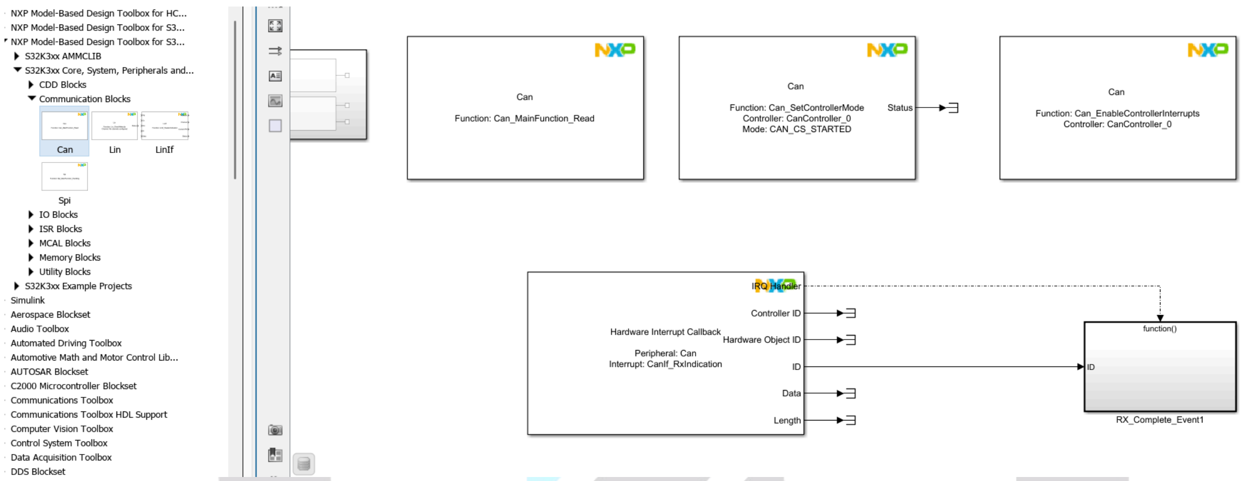


5. You can check in Can Anlyser Tool you will get the data

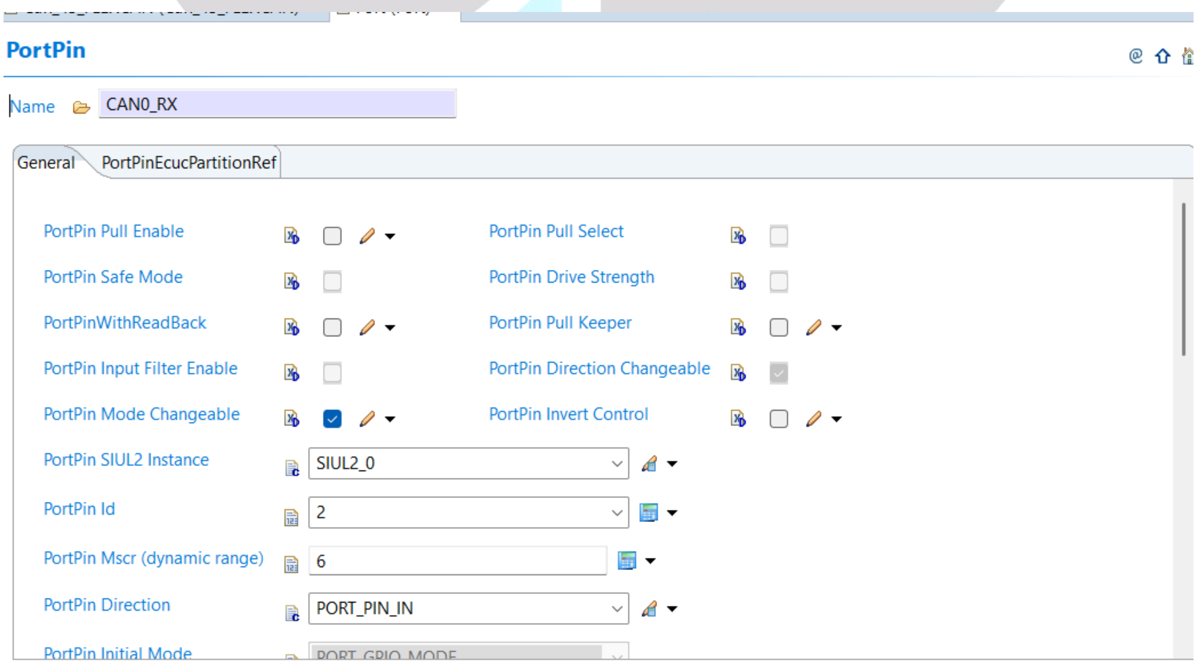


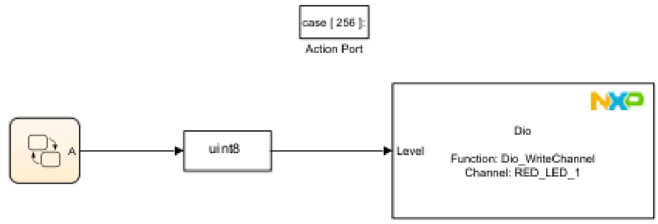
4.2 CAN RECEIVE

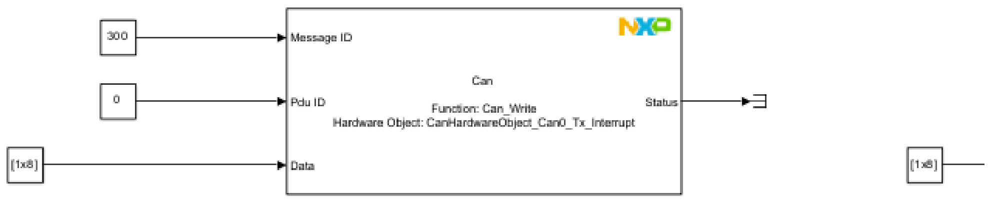
- **Block Selection:** Use the **CAN RECEIVE** block from the library.



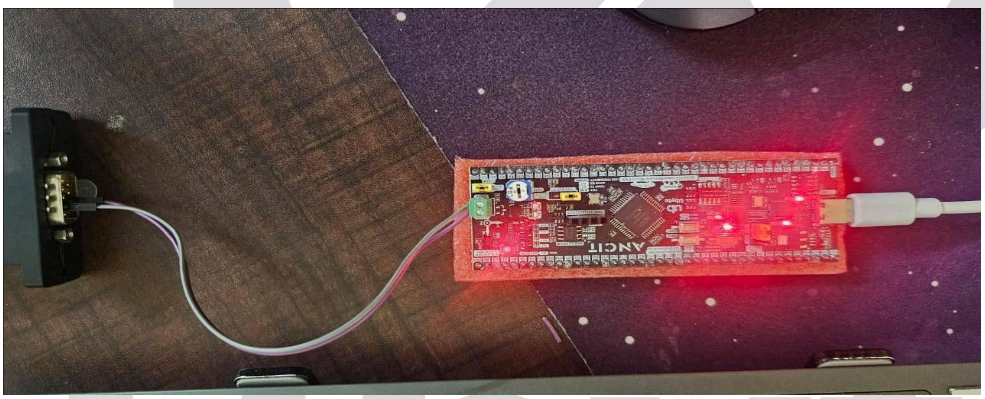
- **Logic Implementation:** Configure the model so that when a specific Receive ID (e.g., 500) is detected, an action occurs (e.g., blinking an LED).

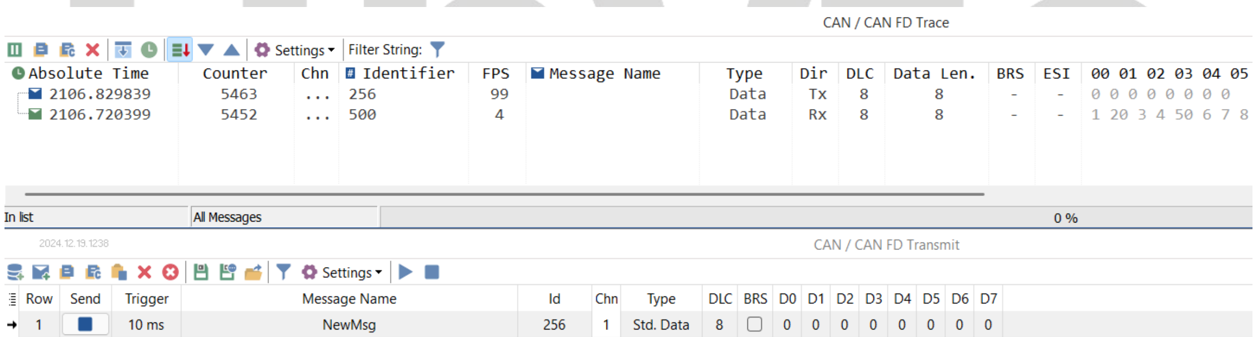






- **Verification:** You should see data on ID 500 in the CAN Analyzer Tool and the RED LED blinking on the board.





CAN / CAN FD Trace

Absolute Time	Counter	Chn	Identifier	FPS	Message Name	Type	Dir	DLC	Data Len.	BRS	ESI	00 01 02 03 04 05
2106.829839	5463	...	256	99		Data	Tx	8	8	-	-	0 0 0 0 0 0 0 0
2106.720399	5452	...	500	4		Data	Rx	8	8	-	-	1 20 3 4 50 6 7 8

In list | All Messages | 0 %

CAN / CAN FD Transmit

Row	Send	Trigger	Message Name	Id	Chn	Type	DLC	BRS	D0	D1	D2	D3	D4	D5	D6	D7
1		10 ms	NewMsg	256	1	Std. Data	8		0	0	0	0	0	0	0	0

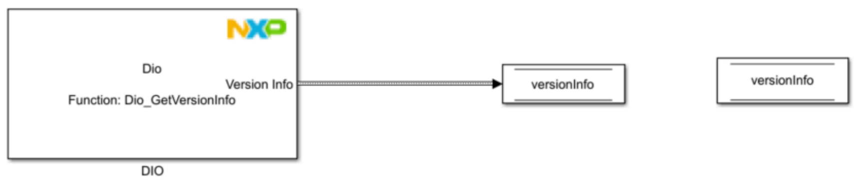
4.3 Digital I/O (DIO) - LED Blinking

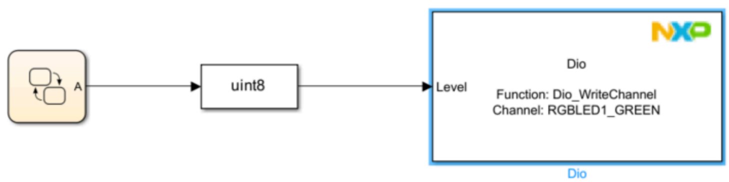
- **Block Setup:** Use the **Dio** block from the library.

- **Configuration:**

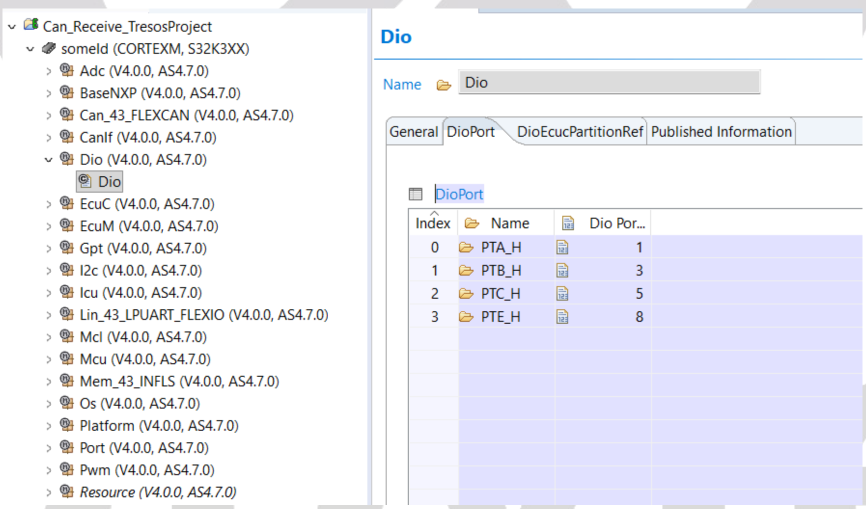
- Press **Configure** to open the setup in EB Tresos.

- Ensure the DIO ports are configured correctly.

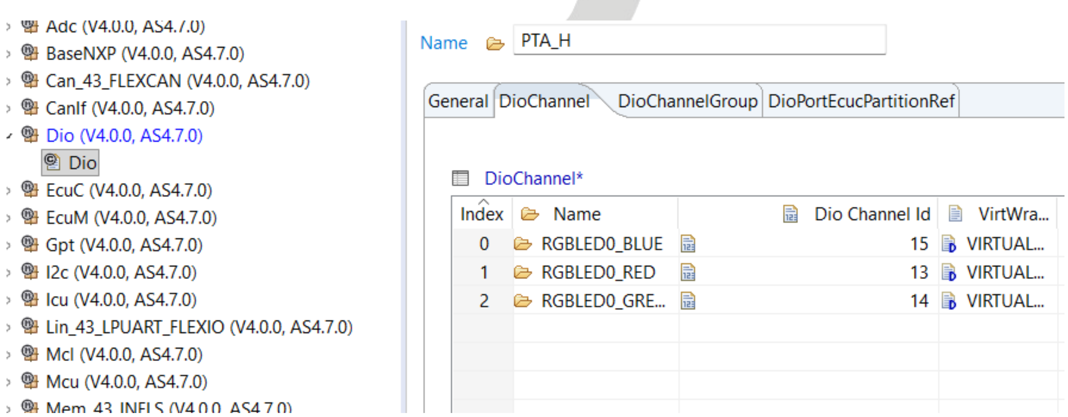




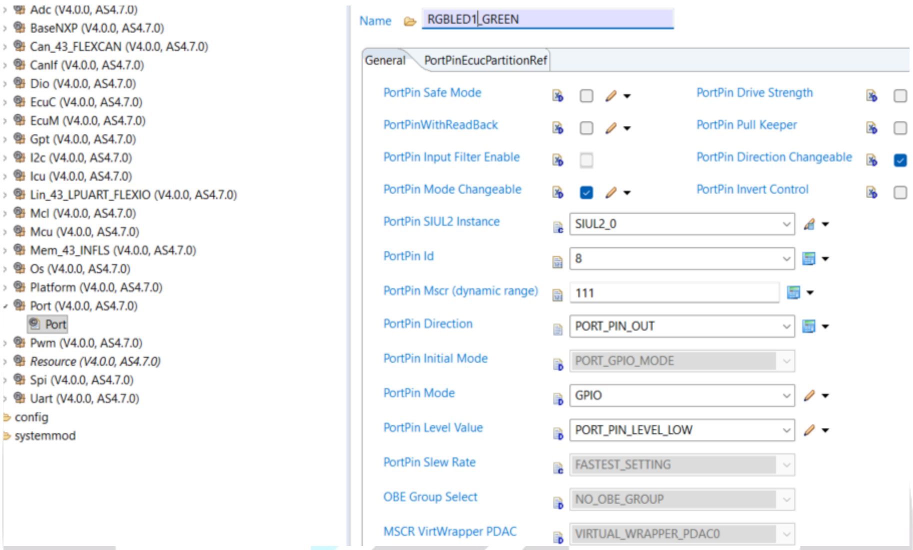
- **Pin Mapping Rule:** Pins 0-15 fall under PT_-L; Pins 16-32 fall under PT_-H.



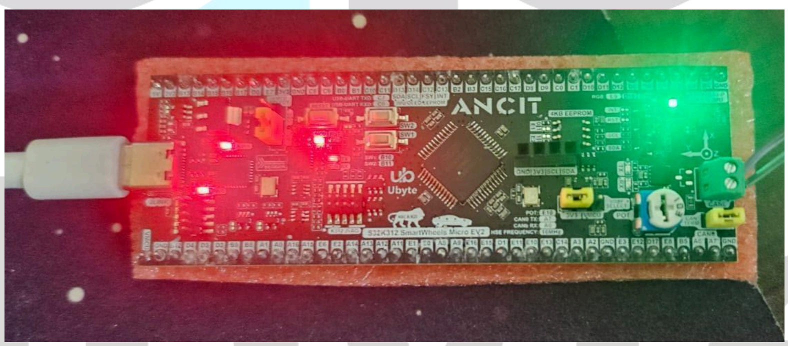
- Select the **Dio channel id** (0-15) based on your pin configuration.



- Navigate to **Port** > **Dio_pins** > **portPin** to finalize pin setup

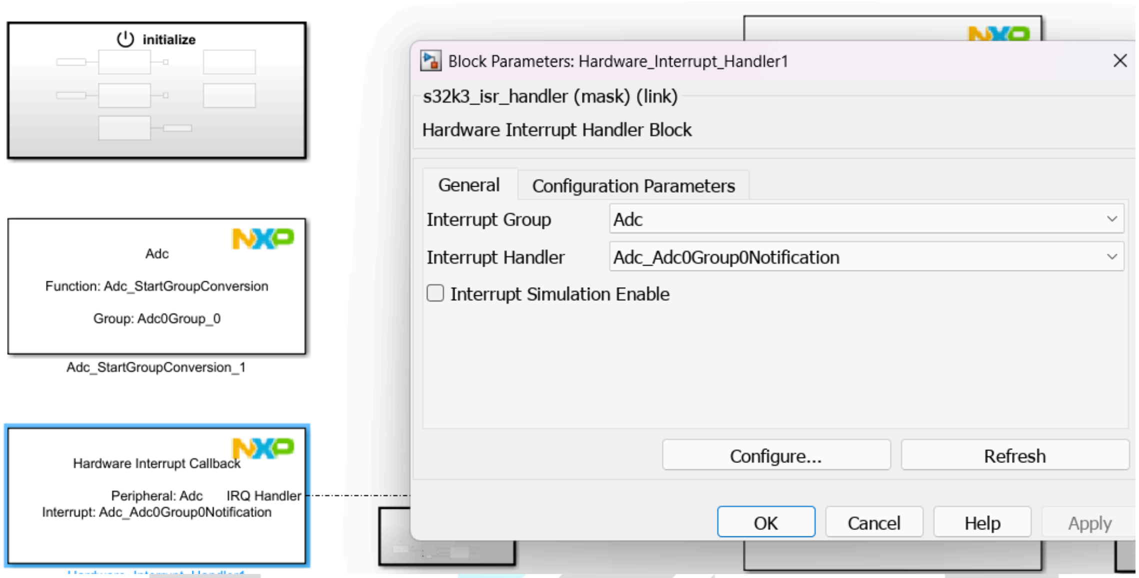


- **Execution:**
- Select the channel (e.g., RGB_LED1_GREEN) in the block settings.
- Update, build, and flash the code to see the RGB_LED1_GREEN blink on the board.

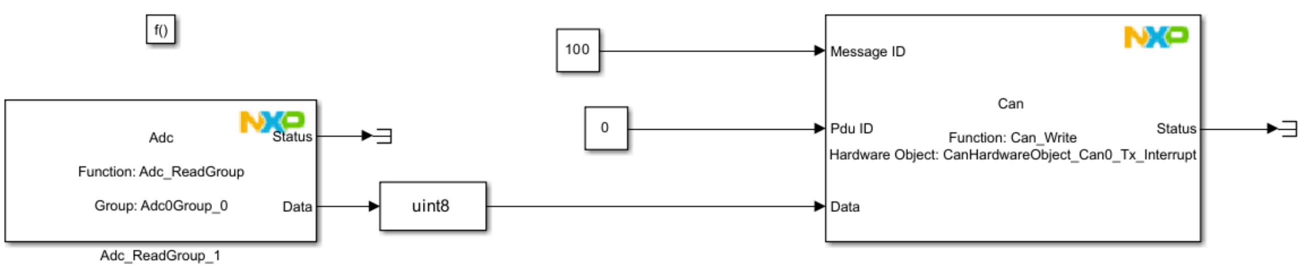


- Analog-to-Digital Converter (ADC)

1. **Block Setup:** Add the **ADC** block to your model.

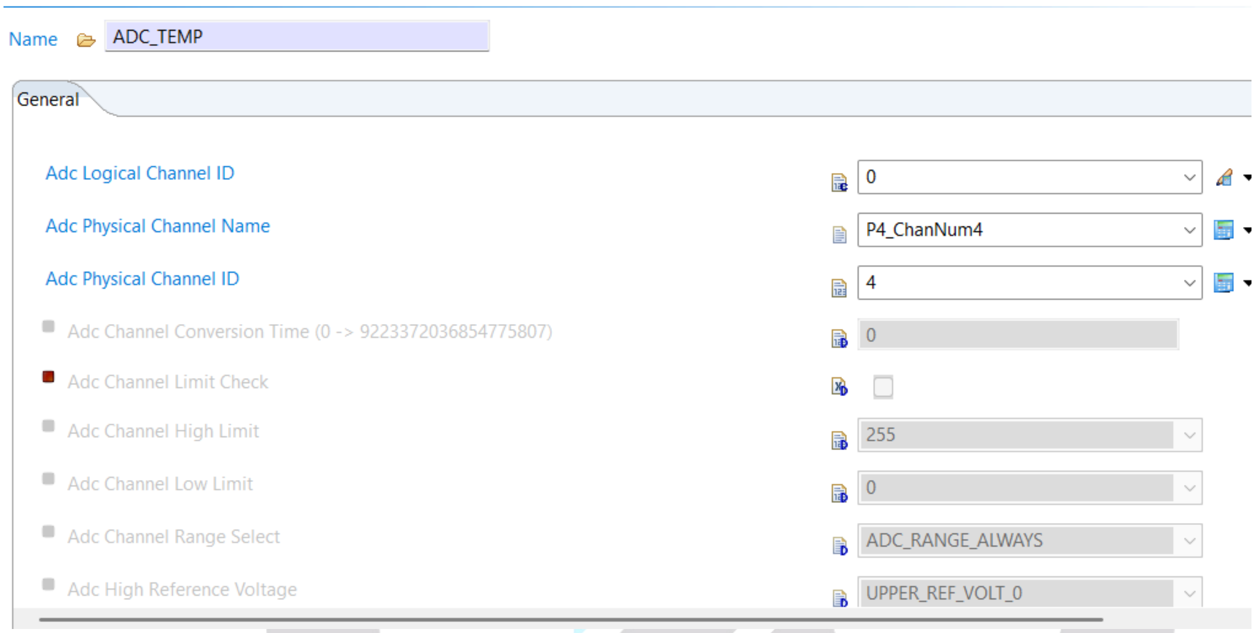


2. **Integration:** Connect the ADC output to a CAN block to visualize ADC value changes on a specific CAN ID (e.g., 100)

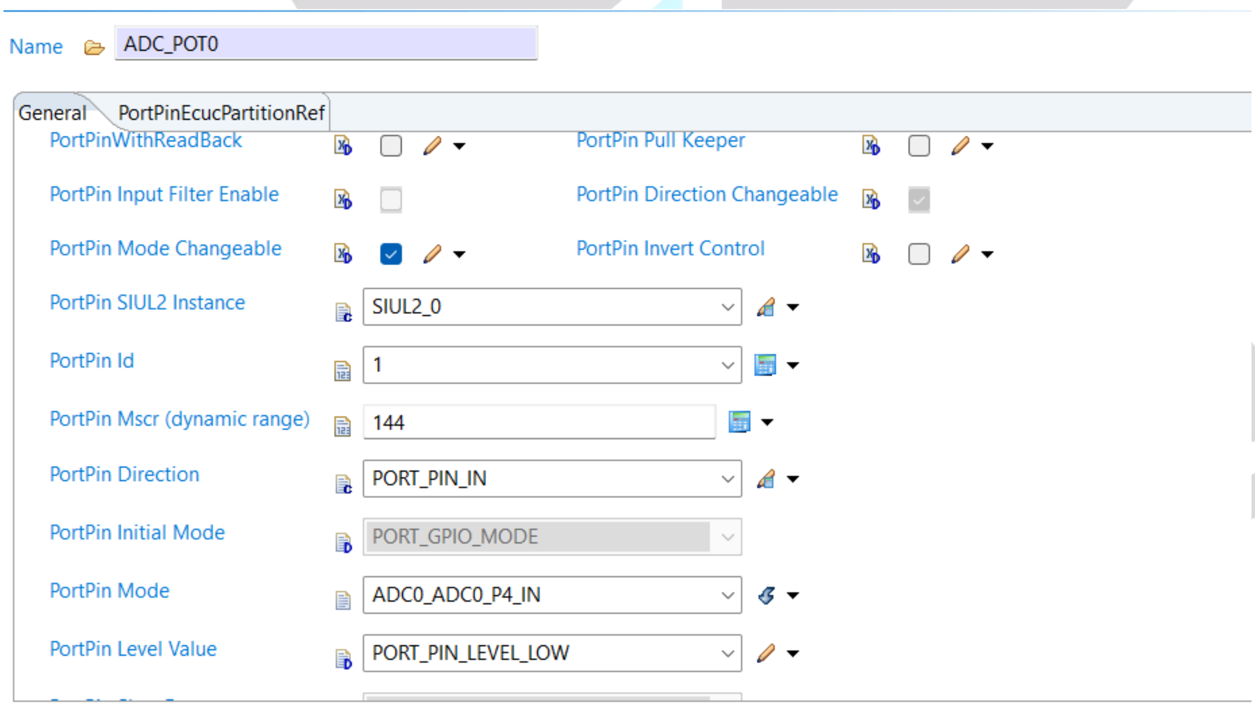


3. **EB Tresos Configuration:**

- Go to **AdcHwUnit** and select the unit (ADC0 or ADC1).
- If using ADC0, go to **Adcchannel** and select the **Adc Physical Channel Name**.



- Navigate to **Port** > **PortContainer** > **ADc pins** and select the correct pin

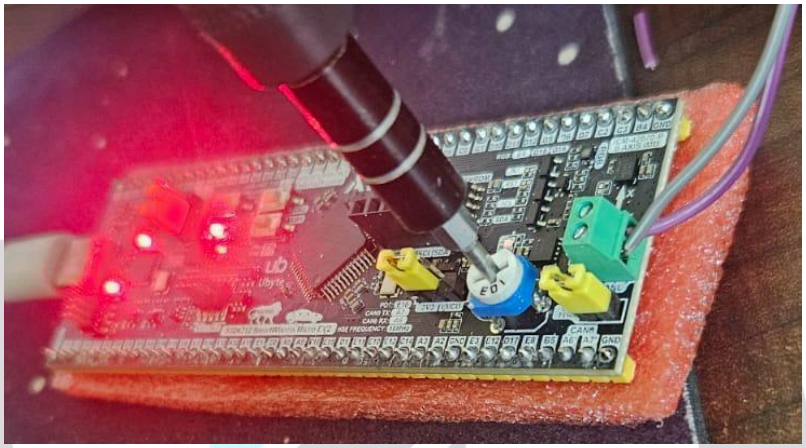


4. **Execution:**

- Update the code, build, and refresh the configuration.
- Select **Adc0** in the Simulink configuration block.

- Run and flash the code.

- Turning the potentiometer on the board will result in changing values visible in the CAN analyzer.



2024.12.19.1238

CAN / CAN FD Trace

Settings ▾ | Filter String:

Absolute Time	Counter	Chn	Identifier	FPS	Mess...	Type	Dir	DLC	Data Len.	BRS	ESI	00 01 02 03 04 05 06
977.132888	9762	...	100	10		Data	Rx	1	1	-	-	189